

High rates of baseline antiretroviral resistance among HIV-infected pregnant women in an HIV referral centre in Rio de Janeiro, Brazil

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Abstract

In order to understand antiretroviral resistance during pregnancy and its impact on HIV vertical transmission, we performed a cross-sectional analysis of 231 HIV-infected pregnant women who fulfilled Brazilian guidelines for antiretroviral testing and had antiretroviral genotypic testing performed between April 2010 and October 2012. At entry into prenatal care, the mean CD4 cell count for this cohort of patients was 406 cells/mm³ (95% CI: 373–438 cells/mm³), while the mean HIV RNA was 24,394 copies/ml (95% CI: 18,275–30,513 copies/ml). Thirty-six women (16%) had detectable antiretroviral-resistant mutations. By 34 weeks gestation, 75% had achieved HIV RNA <400 copies/ml. Our logistic regression model showed the odds of harbouring antiretroviral-resistant virus with a baseline CD4 cell count of <200 cells/mm³ was eight times that of subjects with CD4 cell counts >500 CD4 cells/mm³ (95% CI 1.5–42.73). Six infants were HIV infected, four born to mothers with detectable viraemia at 34 weeks and two born to mothers who were lost to follow up. Antiretroviral resistance is common in prenatal care but did not increase vertical transmission if viral load was appropriately suppressed. Genotyping should be considered in Brazil in order to assist initiation of appropriate combination antiretroviral therapy during pregnancy to suppress viral load to avoid vertical transmission.

Keywords

HIV, AIDS, women, treatment, antiretroviral therapy, antiretroviral resistance during pregnancy, vertical transmission of HIV, prevention of mother to child transmission, PMTCT

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Introduction

Successful prevention of HIV perinatal transmission involves the use of potent combination antiretroviral therapy (cART) starting before pregnancy and continued during pregnancy, along with antiretroviral (ARV) as post-exposure prophylaxis to the infant.^{1–3} The potential acquisition or detection of maternal ARV-resistant strains prior to or during pregnancy is an area of concern, as it could lead to failure of prevention of mother to child transmission (PMTCT) interventions with transmission of resistant HIV-1 strains to HIV-exposed infants.⁴ To circumvent potential transmission of resistant strains, routine ARV susceptibility testing is recommended in the United States to all women initiating cART during pregnancy, but this is not routinely performed in Brazil.^{5,6}

Brazil has provided universal access to cART to HIV/AIDS-infected patients since 1996 and to all HIV-infected pregnant patients in the country since 2004 regardless of

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CD4 cell count. Because there are limited data on HIV resistance patterns among pregnant HIV-infected patients, this present study aims to characterise the frequency and type of ARV resistance mutations among HIV-infected pregnant women seeing PMTCT services at a tertiary, public hospital in Rio de Janeiro, Brazil. Potential risk factors often associated with the presence or absence of ARV resistance in the patient population were evaluated, as well as incidence of vertical transmission.

Methods

Study design, population and enrollment criteria

This is a cross-sectional study analysis that was conducted at an HIV PMTCT referral centre (Hospital Federal dos Servidores do Estado – HFSE) in Rio de Janeiro in a cohort of HIV-infected pregnant women initiating prenatal care at HFSE between April 2010 and October 2012. All HIV-infected pregnant women referred to the institution used cART according to Brazilian and US guidelines.^{7,8} Subjects willing and able to provide informed consent were included in the study, if they were presented with a baseline HIV RNA viral load that met criteria for susceptibility testing according to Brazilian Ministry of Health guidelines (recommended for patients with >2000 copies of HIV RNA/ml of plasma until September 2011; and >1000 copies of HIV RNA/ml of plasma thereafter).

TRUGENE HIV-1 DNA sequencing platform (Visible Genetics, Canada) was used for genotypic

analysis of patient viral isolates and characterisation of HIV-1 subtypes.⁹ Relevant HIV-1 ARV genotypic mutations were determined using the Stanford database (<http://hivdb.stanford.edu>). The HIVdb algorithm was used to assess drug resistance mutations for both naïve and ART-exposed subjects. Only individual mutation data were available for this analysis. The study was approved by the institutional review board.

Statistical analysis

Statistical analysis was performed using SAS 9.2 (SAS Institute, Inc., Cary, North Carolina). Pearson's chi square test, comparing resistant and non-resistant iso-types by drug type, was run for categorical variables and applied Yates's Correction as necessary. Multivariable logistic regression was performed for evaluation of risk factors independently associated with genotypic resistance. Variables included in multivariate model included race, marital status, income, HIV diagnosis prior to pregnancy, cART use, prior sexually transmitted infection (STI) diagnosis, current drug use and CD4 count at entry.

Results

Study population demographics

As seen in Figure 1, 448 HIV-infected pregnant women received prenatal care at the HFSE institution between 4/2010 and 10/2012. Within this population, 232 (52%)

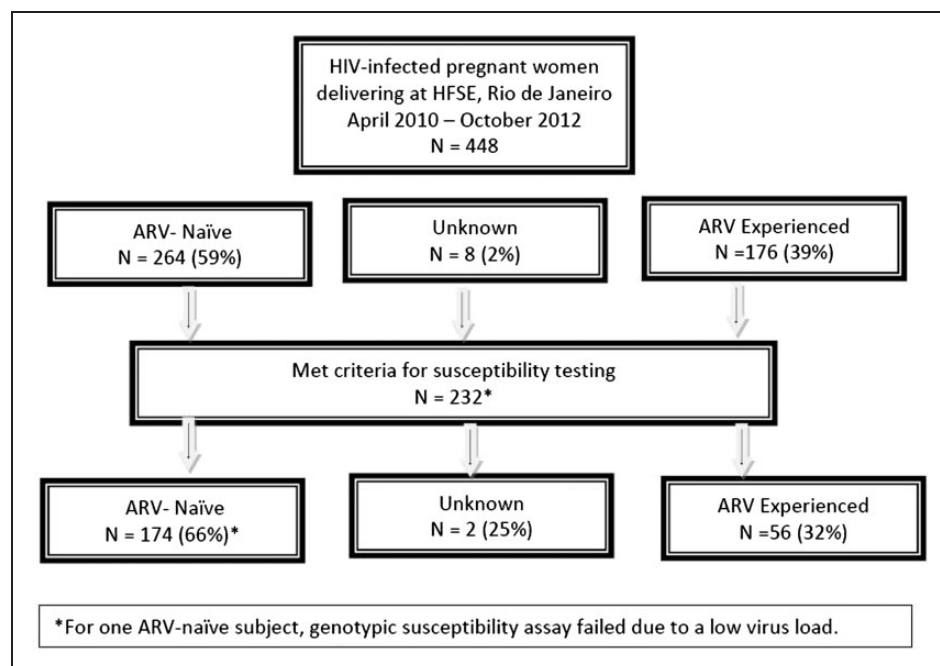


Figure 1. Flow diagram of cohort.

Table 1. Patient demographics, details of ARV use (N = 231).

	N (%)
Age at entry (N = 231)	
Under 18	16 (6.9)
18–30	147 (63.6)
Older than 30	68 (29.4)
HIV diagnosed before pregnancy	105 (54.4)
Family income	
Impoverished (under \$560 month ≤2 minimum wages)	167 (79.1)
Race (N = 231)	
African origin	180 (77.9)
Caucasian	51 (22.1)
Marital status – single	75 (32.5)
Years of education (N = 229)	
Incomplete grade school	105 (45.9)
Incomplete high school	69 (30.1)
High school graduate	55 (24)
Uses tobacco	161 (70.9)
Consumes alcohol	93 (41)
Uses Illicit drug use	35 (15)
Previously diagnosed STI	65 (28)
CD4 count at entry (cells per μ L) (N = 230)	
0–200	46 (20)
201–499	118 (51.3)
500+	66 (28.7)
HIV viral load at entry (copies/ml) (N = 229)	
Undetectable <400	0
Detectable under 1000	2 (0.9)
Detectable under 10,000	116 (50.7)
Detectable under 100,000	98 (42.8)
Greater than 100,000	13 (5.6)
HIV viral load at 34 weeks gestation (copies/ml) (N = 195)	
Undetectable <400	147 (75.4)
400+	48 (24.6)
HIV Subtype (N = 172)	
A1	1 (0.6)
B	146 (84.9)
C	4 (2.3)
F	2 (1.2)
FI	16 (9.3)
B/F	3 (1.7)
cART use before genotyping (N = 229)	
cART naïve	173 (75.6)
cART previous use	56 (24.5)
Details of cART use (N = 231)	
cART naïve	173 (74.9)
No information	2 (0.9)
HIV PMTCT prophylaxis	16 (6.9)
Treatment	40 (17.3)

(continued)

Table 1. Continued.

	N (%)
Gestational age when cART initiated (N = 219)	
First trimester	21 (9.6)
Second trimester	140 (63.9)
Third trimester	58 (26.5)
Gestational age at genotyping (N = 228)	
First trimester	33 (14.5)
Second trimester	137 (60.1)
Third trimester	58 (25.4)
ARV resistance (N = 231)	
0 classes of drugs	195 (84.4)
1 class of drugs	18 (7.8)
1 + classes of drugs	18 (7.8)

fulfilled Ministry of Health criteria for HIV genotypic susceptibility testing and 231 had plasma specimens assayed for this purpose. Table 1 demonstrates the breakdown of the demographics and characteristics of our study population. Sixty-four percent of women were between 18 and 30 years of age, and over half of the cohort had been diagnosed before the current pregnancy. The vast majority of patients fulfilled Brazilian criteria for low socio-economic status and were of African heritage, which mirrors the social profile of the HIV epidemic in pregnant women in Rio de Janeiro. Forty-six percent of patients had not completed primary school and 71% used tobacco during pregnancy. Among 231 eligible subjects, 105 patients were diagnosed with HIV infection prior to entering prenatal care at HFSE, and of these, 56 had previous exposure to cART. This includes three women who were perinatally infected themselves and had been on cART for many years. The mean CD4 cell count at study entry was 406 cells/mm³ (95% CI: 373–438 cells/mm³) and the mean viral load 24,394 copies/ml (95% CI: 18,275–30,513 copies/ml). The majority of women initiated cART during the second trimester of pregnancy, the most common first line regimen being zidovudine (ZDV) + lamivudine (3TC) + lopinavir/ritonavir (LPV/RTV) (*n* = 175). By 34 weeks gestation, 147/231 women (64%) achieved an undetectable viral load (<400 copies/ml); 48/231 subjects (20.8%) had levels above 400 copies/ml plasma and 36/231 subjects (16%) did not have a viral load result at 34 weeks, either because they did not come for that visit or because of premature delivery.

ARV genotypic mutations

Genotypic testing was performed on 173 cART-naïve subjects, 56 cART-experienced subjects, and two subjects with unknown previous cART experience.

Thirty-six women (15.6% of 231 subjects) had at least one ARV genotypic resistance mutation detected, as seen in Table 2.

As seen in Table 3, the presence of detectable nucleoside reverse transcriptase inhibitor (NRTI)-resistant HIV-1 isolates was 15.5 times more frequent among cART-experienced subjects as compared to

subjects who were cART-naïve (95% CI: 4.18, 57.16), i.e. 12 of 56 (21%) women with prior ART exposure harboured NRTI-resistant virus vs. 3 of 173 (1.7%) women without prior ART use. In the NRTI-resistant population, the average number of specific genetic mutations leading to resistance was 1.6. The most common mutation was M184V/L with 31% of patients (11/36) patients expressing this genotype.

cART-experienced women had 3.5 times the likelihood of harbouring non-nucleoside reverse transcriptase inhibitor (NNRTI)-resistant HIV as compared to ARV-naïve women (95% CI: 1.57, 7.85), i.e., 14 of 56 (25%) subjects with prior NNRTI exposure vs. 15 of 173 (8.7%) patients without prior ART exposure. Patients with NNRTI-resistant virus had on average one resistance mutation, the most common being the K103N in 14 patients.

Table 2. ARV genotypic mutations identified in 36/231 subjects (16%).

Genotypic resistance mutations identified	ARV-naïve patients (N = 17)	ARV-experienced patients (N = 19)	Total
K70R	0	4	4
K219N	0	1	1
L210W	0	1	1
M184V/L	1	10	11
M41L	1	1	2
T69Na	1	3	4
T215Y/F	0	1	1
No. of patients with any NRTI mutation	3	12	15
A98G	0	2	2
F227C ^a	1	0	1
G190A/E/Q/S	2	1	3
K101E	0	1	1
K103N	7	7	14
V108I	0	1	1
V179D/E/F ^a	3	1	4
Y181C	2	1	3
Y188L	1	1	2
No. of patients with any NNRTI mutation	15	14	29
I50L	0	1	1
L90M	0	1	1
L10F/I/R/V ^a	1	3	4
L76V	0	1	1
M36I/L/V and/or A71I/L/T/V and/or V77I	0	3	3
M46I/L	1	2	3
V82A/S/F/T	0	3	3
No. of patients with any PI mutation	1	4	5
Total no. of resistance mutations identified ^a	21	50	71

ARV: antiretroviral, NRTI: nucleoside reverse transcriptase inhibitor, NNRTI: non-nucleoside reverse transcriptase inhibitor, PI: protease inhibitor.

^aResistance conferred by these specific mutations in ARV-naïve patients is not well characterised.

Table 3. Odds ratios for the effect of previous ART use on resistance (N = 231).

NNRTI resistance			
	NNRTI resistance	No resistance	Total
Prior ARV use	14	42	56
ARV naïve	15	158	173
Missing, n = 2			229
Prior ARV use	Crude OR	95% confidence interval	
Yes	3.51*	1.57, 7.85	
No	1	–	
NRTI resistance			
	NRTI resistance	No resistance	Total
Prior ARV use	12	44	56
ARV naïve	3	170	173
Missing, n = 2			229
Prior ARV use	Crude OR	95% confidence interval	
Yes	15.45*	4.18, 57.16	
No	1	–	
Any resistance (NNRTI, NRTI, PI)			
	Any resistance	No resistance	Total
Prior ARV use	19	37	56
ARV naïve	17	156	173
Missing, n = 2			229
Prior ARV use	Crude OR	95% confidence interval	
Yes	4.71*	2.23, 9.93	
No	1	–	

ARV: antiretroviral; NRTI: nucleoside reverse transcriptase inhibitor; NNRTI: non-nucleoside reverse transcriptase inhibitor; PI: protease inhibitor.

*p < 0.05.

Protease inhibitor (PI) resistance was relatively uncommon, present only in five (2%) individuals. Due to the small number of PI-resistant mutations encountered in this cohort, we did not include an odds ratio in our analyses. The average number of genotypic mutations leading to resistance in the PI-resistant patient population was 3.2.

Multivariate analysis

Our multivariate analysis showed that income, exposure to STIs, drug use, cART use and marital status did not appear to be associated with ARV resistance. However, the odds of harbouring virus with any ARV resistance was eight times more among subjects with a baseline CD4 cell count <200 cells/mm³ as compared to subjects with CD4 counts >500 cells/mm³ (95% CI: 1.5, 42.73, $p < 0.01$). (Data not shown).

Effect of CD4+ cell count on specific genotypic resistance

As CD4 cell count was associated with ARV resistance in our model, we decided to further examine this relationship for each type of drug. By means of Pearson's chi square analysis using categories defined in Table 1, we found that NNRTI resistance was influenced by baseline CD4 cell count (p : 0.0437). This interaction was not evident among patients harbouring virus with ARV resistance to other classes of ARVs (p : 0.1134; p : 0.2775).

ARV resistance and transmission

Six neonates of 231 women were HIV-infected (2.3% perinatal transmission rate). All infected infants were born to cART-naïve women, four of them presenting to care after 27 weeks of gestation, and two additional patients lost to follow up, abandoning treatment and returning with infected infants following delivery. No transmitting mothers harboured detectable ARV-resistant mutations; five mothers had subtype B virus, and one had subtype F1. Transmitting mothers had a relatively high HIV viral load at entry into prenatal care (mean VL: 42,516 cps/ml, range: 1949–113,424 copies/ml). Four of these subjects still had detectable virus at the time of delivery and two did not return for a repeat viral load at 34 weeks of gestation.

Discussion

To accomplish the goal of creating an HIV-free generation of 2015, we must offer pregnant women a comprehensive plan to prevent HIV transmission to infants.¹⁰ If viral isolates with genotypic mutations

conferring resistance are vertically transmitted, treatment options available for neonates are further reduced.^{11,12} Many populations still present with moderate levels of transmitted drug resistance (5–15% prevalence) as defined by the WHO HIV Drug Resistance Report 2012. This study's ARV-naïve patient population fits within this range with 9.2% of subjects harbouring virus with at least one class of ARV resistance, similar to recent data from Rio de Janeiro and throughout the rest of Brazil and Latin America.^{13–15} Secondary ARV resistance was significantly higher than primary resistance, with 32.1% of cART-experienced women in the cohort infected with resistance isolates.^{16,17}

Among the NRTI-resistant isolates, M184V/L was the most common in our cohort. We also noted K103N mutations leading to resistance against specific NNRTIs. These two mutations have been well characterised in the HIV-infected Brazilian population and are likely associated with the early high use of nevirapine-based cART regimens for PMTCT purposes in southeastern and southern Brazil, from 1998 to 2001.^{18–20} Although the Stanford algorithm employed in this study for characterisation of resistant mutations among ARV-naïve subjects might tend to overestimate resistance, removal of questionable resistance mutations from the analysis did not change the number of naïve subjects harbouring resistance to any class of ARVs. The moderate prevalence of ARV resistance coupled with the high frequency of the aforementioned mutations which abrogate many commonly-employed treatment plans, enforce the necessity to promote routine and universal ARV susceptibility testing to all Brazilian HIV-infected pregnant women for PMTCT purposes.

Although high rates of baseline ARV resistance during pregnancy is of significant concern, early initiation of ARV therapy appears to play the most important role in preventing mother to child HIV transmission. In the present cohort, 75% of subjects with available results at 34 weeks had achieved undetectable viral loads (<400 copies/ml). All six infants who acquired HIV infection were born to ARV-naïve women with detectable or unknown viral loads at delivery. Women who received cART in the first two trimesters of pregnancy, even if they had resistant viral strains, did not vertically transmit HIV to their infants. Among the 58 women who initiated treatment in the third trimester of pregnancy, 4 (7%) transmitted HIV to their infants. These results support the need for more intensive viral load monitoring and earlier initiation of cART among pregnant women as critical components of a successful PMTCT policy.

Previous studies have suggested that higher baseline CD4 cell counts may be associated with lower rates of

general ARV resistance, although other studies negate this association.^{21,22} In the present analysis, after adjusting for other variables in the multivariate model, individuals with a CD4 cell count greater than 500 cells/mm³ had eight times less risk of ARV-resistant virus as compared to those with CD4 cell counts below this threshold. Subjects with lower CD4 cell counts were more likely to have received prior ARVs, have longer duration of infection, and have had more time for development of resistance mutations and acquisition of resistance in the past. Three women enrolled in this analysis were perinatally-infected subjects themselves, having prolonged ARV exposure and a number of ARV resistance mutations. The results of our multivariate analysis demonstrating no association between resistance and cART are likely a result of small numbers especially since a low CD4 count is a confounding variable.

There is a paucity of data regarding ARV resistance in the pregnant Brazilian population, particularly when ARVs are used for HIV PMTCT purposes primarily. Aside from describing the prevalence of resistance mutations in our cohort, we explored potential causal factors associated with multiple forms of drug resistance. Our study design was constrained by assay limitations and country guidelines, as 216 individuals with a low viral load often did not have enough copy numbers to amplify resistance mutations during genotypic testing. Given the high rates of resistance seen in both cART-experienced and cART-naïve patients, future studies of ARV resistance in Brazil should include all subjects with detectable viraemia at baseline in order to further characterise the role of ARV resistance to guide HIV PMTCT management.

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Declaration of Conflicting Interests

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